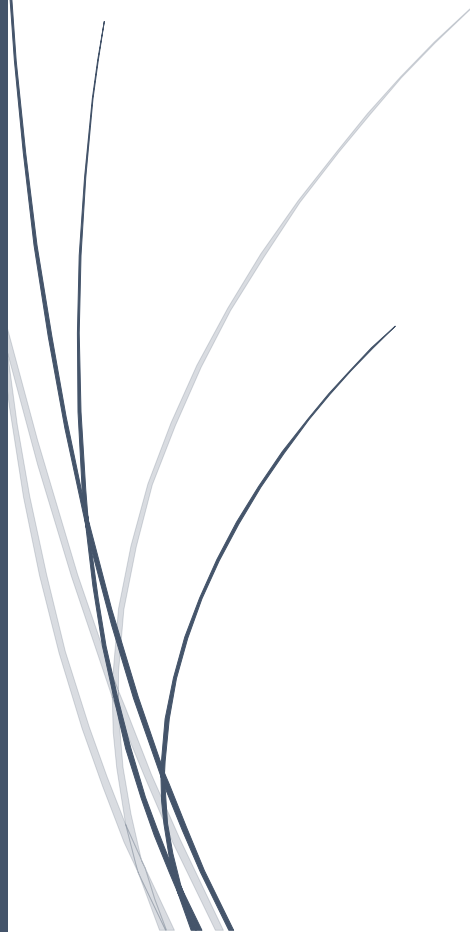


A dark blue vertical bar runs down the left side of the page. A blue arrow points to the right from the bar, containing the text 'Mr Michael Weiss'.

Mr Michael Weiss

Operating System

Proof of Concept Linux virtual
network Project

Several thin, curved lines in dark blue and light grey originate from the bottom left and sweep upwards and to the right.

Marco dos Santos

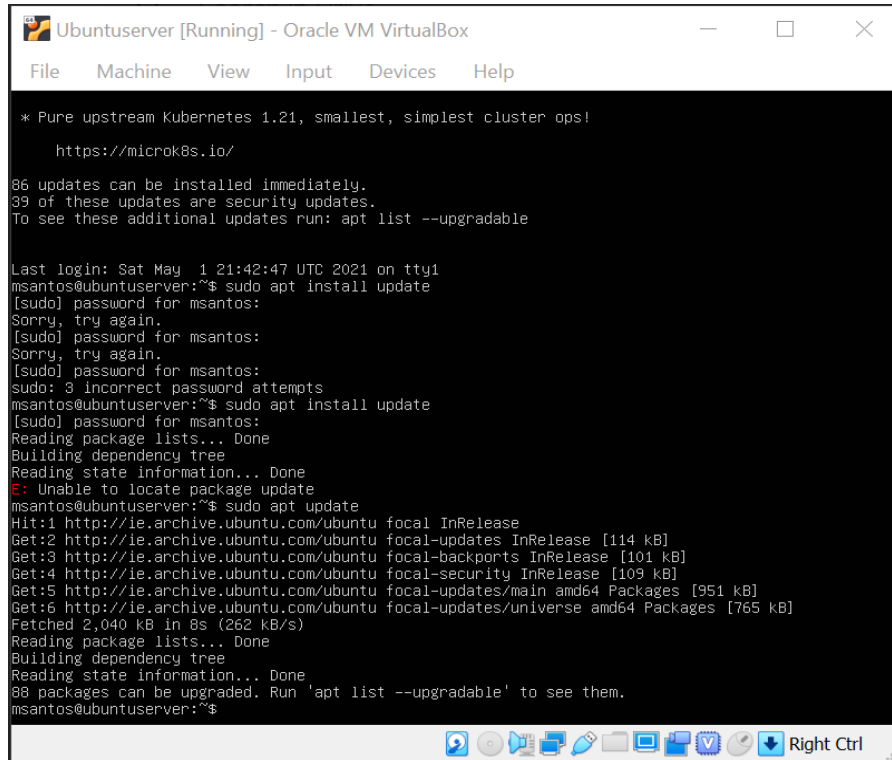
STUDENT NUMBER: 2020333

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Obtain Linux updated and ping to the servers

Ubuntuserver update



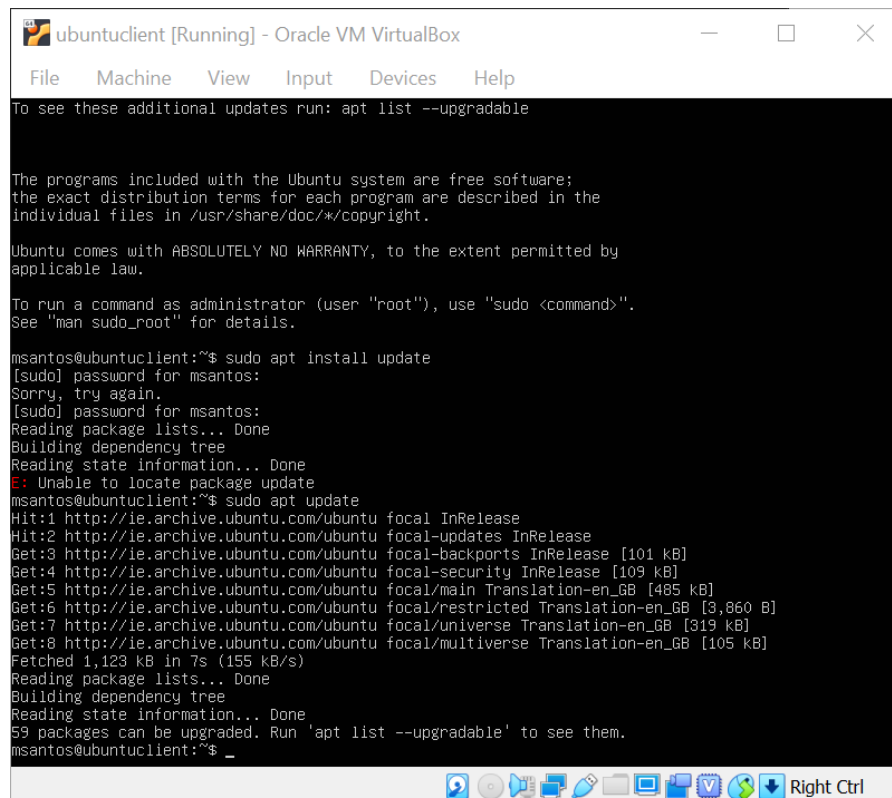
```
* Pure upstream Kubernetes 1.21, smallest, simplest cluster ops!

https://microk8s.io/

86 updates can be installed immediately.
39 of these updates are security updates.
To see these additional updates run: apt list --upgradable

Last login: Sat May 1 21:42:47 UTC 2021 on tty1
msantos@ubuntuserver:~$ sudo apt install update
[sudo] password for msantos:
Sorry, try again.
[sudo] password for msantos:
Sorry, try again.
[sudo] password for msantos:
sudo: 3 incorrect password attempts
msantos@ubuntuserver:~$ sudo apt install update
[sudo] password for msantos:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package update
msantos@ubuntuserver:~$ sudo apt update
Hit:1 http://ie.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://ie.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Get:3 http://ie.archive.ubuntu.com/ubuntu focal-backports InRelease [101 kB]
Get:4 http://ie.archive.ubuntu.com/ubuntu focal-security InRelease [109 kB]
Get:5 http://ie.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [951 kB]
Get:6 http://ie.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [765 kB]
Fetched 2,040 kB in 8s (262 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
88 packages can be upgraded. Run 'apt list --upgradable' to see them.
msantos@ubuntuserver:~$
```

Ubuntuclient update



```
To see these additional updates run: apt list --upgradable

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

msantos@ubuntuclient:~$ sudo apt install update
[sudo] password for msantos:
Sorry, try again.
[sudo] password for msantos:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package update
msantos@ubuntuclient:~$ sudo apt update
Hit:1 http://ie.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://ie.archive.ubuntu.com/ubuntu focal-updates InRelease
Get:3 http://ie.archive.ubuntu.com/ubuntu focal-backports InRelease [101 kB]
Get:4 http://ie.archive.ubuntu.com/ubuntu focal-security InRelease [109 kB]
Get:5 http://ie.archive.ubuntu.com/ubuntu focal/main Translation-en_GB [485 kB]
Get:6 http://ie.archive.ubuntu.com/ubuntu focal/restricted Translation-en_GB [3,860 B]
Get:7 http://ie.archive.ubuntu.com/ubuntu focal/universe Translation-en_GB [319 kB]
Get:8 http://ie.archive.ubuntu.com/ubuntu focal/multiverse Translation-en_GB [105 kB]
Fetched 1,123 kB in 7s (155 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
59 packages can be upgraded. Run 'apt list --upgradable' to see them.
msantos@ubuntuclient:~$
```

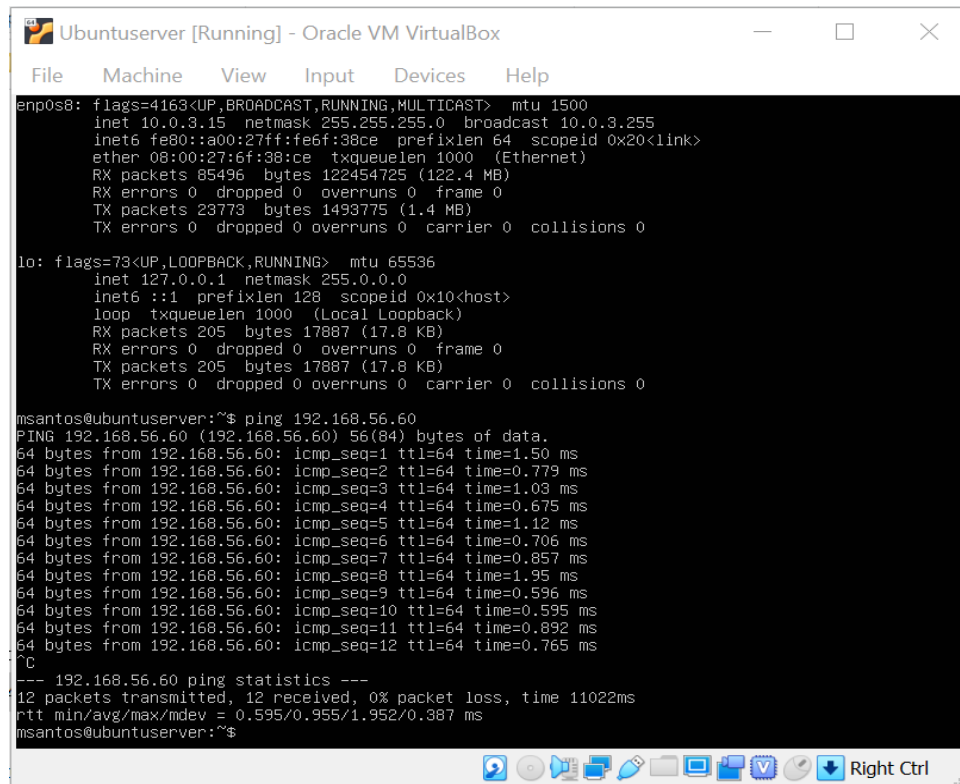
Ubuntuserver upgrade

```
E: Unable to locate package upgrade
msantos@ubuntuserver:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.4.0-65 linux-headers-5.4.0-65-generic linux-image-5.4.0-65-generic
  linux-modules-5.4.0-65-generic linux-modules-extra-5.4.0-65-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libbevdv2 libimobiledevice6 libplist3 libupower-glib3 libusbmuxd6 linux-headers-5.4.0-72
  linux-headers-5.4.0-72-generic linux-image-5.4.0-72-generic linux-modules-5.4.0-72-generic
  linux-modules-extra-5.4.0-72-generic upower usbmuxd
The following packages will be upgraded:
  alsa-ucm-conf apport apt apt-utils bind9-dnsutils bind9-host bind9-libs ca-certificates
  cloud-init curl dirmngr distro-info-data friendly-recovery git git-man gnupg gnupg-l10n
  gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv grub-common grub-pc
  grub-pc-bin grub2-common initramfs-tools initramfs-tools-bin initramfs-tools-core
  isc-dhcp-client isc-dhcp-common landscape-common libapt-pkg6.0 libasound2 libasound2-data
  libcurl3-gnutls libcurl4 libglib2.0-0 libglib2.0-bin libglib2.0-data libhogweed5 libldap-2.4-2
  libldap-common libnettle7 libnss-systemd libpam-systemd libpci3 libprocps8 libpython3.8
  libpython3.8-minimal libpython3.8-stdlib libseccomp2 libssl1.1 libsystemd0 libudev1 libxmlb1
  libzstd1 linux-firmware linux-generic linux-headers-generic linux-image-generic open-iscsi
  open-vm-tools openssh-client openssh-server openssh-sftp-server openssl pciutils pollinate
  procps python3-apport python3-distupgrade python3-problem-report python3-software-properties
  python3-twisted python3-twisted-bin python3-update-manager python3.8 python3.8-minimal screen
  snapd software-properties-common sosreport systemd systemd-sysv systemd-timesyncd thermald tmux
  ubuntu-keyring ubuntu-release-upgrader-core udev update-manager-core update-notifier-common
96 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 248 MB of archives.
After this operation, 382 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Ubuntuclient upgrade

```
E: Unable to locate package upgrade
msantos@ubuntuclient:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.4.0-65 linux-headers-5.4.0-65-generic linux-image-5.4.0-65-generic
  linux-modules-5.4.0-65-generic linux-modules-extra-5.4.0-65-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libbevdv2 libimobiledevice6 libplist3 libupower-glib3 libusbmuxd6 linux-headers-5.4.0-72
  linux-headers-5.4.0-72-generic linux-image-5.4.0-72-generic linux-modules-5.4.0-72-generic
  linux-modules-extra-5.4.0-72-generic upower usbmuxd
The following packages will be upgraded:
  alsa-ucm-conf apport apt apt-utils bind9-dnsutils bind9-host bind9-libs ca-certificates
  cloud-init curl dirmngr distro-info-data friendly-recovery git git-man gnupg gnupg-l10n
  gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv grub-common grub-pc
  grub-pc-bin grub2-common initramfs-tools initramfs-tools-bin initramfs-tools-core
  isc-dhcp-client isc-dhcp-common landscape-common libapt-pkg6.0 libasound2 libasound2-data
  libcurl3-gnutls libcurl4 libglib2.0-0 libglib2.0-bin libglib2.0-data libhogweed5 libldap-2.4-2
  libldap-common libnettle7 libnss-systemd libpam-systemd libpci3 libprocps8 libpython3.8
  libpython3.8-minimal libpython3.8-stdlib libseccomp2 libssl1.1 libsystemd0 libudev1 libxmlb1
  libzstd1 linux-firmware linux-generic linux-headers-generic linux-image-generic open-iscsi
  open-vm-tools openssh-client openssl pciutils pollinate procps python3-apport
  python3-distupgrade python3-problem-report python3-software-properties python3-twisted
  python3-twisted-bin python3-update-manager python3.8 python3.8-minimal screen snapd
  software-properties-common sosreport systemd systemd-sysv systemd-timesyncd thermald tmux
  ubuntu-keyring ubuntu-release-upgrader-core udev update-manager-core update-notifier-common
94 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 247 MB of archives.
After this operation, 382 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

UbuntuServer ping to the UbuntuClient



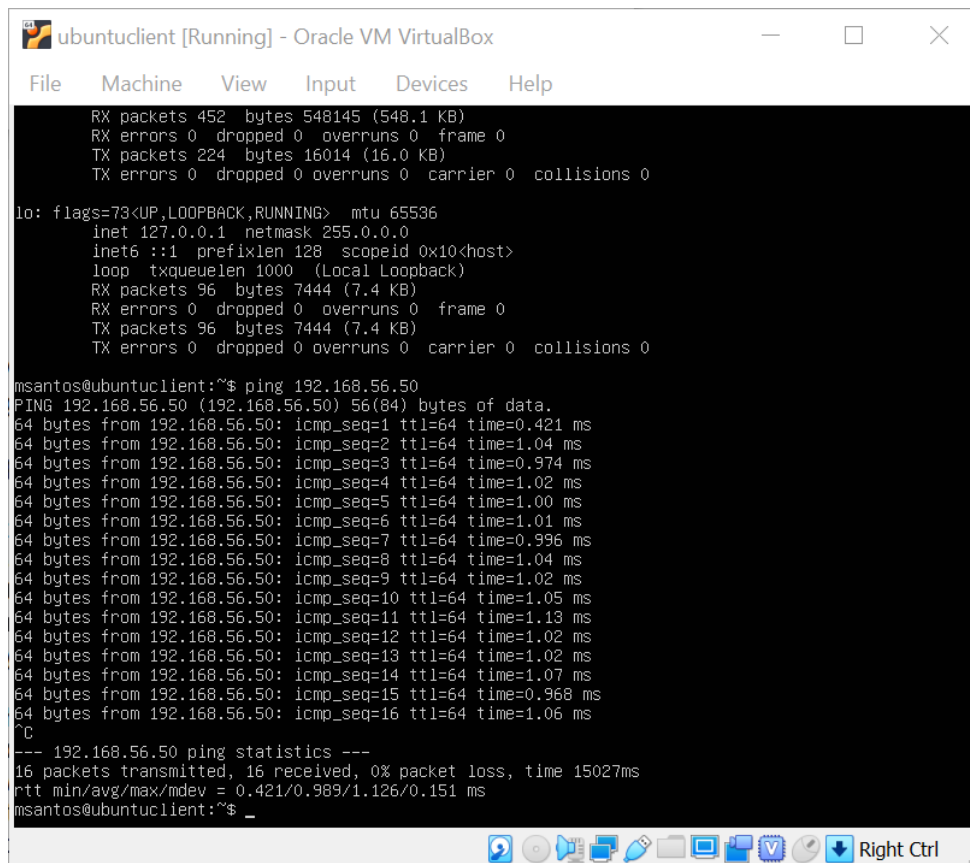
```
UbuntuServer [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.3.15 netmask 255.255.255.0 broadcast 10.0.3.255
    inet6 fe80::a00:27ff:fe6f:38ce prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:6f:38:ce txqueuelen 1000 (Ethernet)
    RX packets 85496 bytes 122454725 (122.4 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 23773 bytes 1493775 (1.4 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 205 bytes 17887 (17.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 205 bytes 17887 (17.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msantos@ubuntuServer:~$ ping 192.168.56.60
PING 192.168.56.60 (192.168.56.60) 56(84) bytes of data:
64 bytes from 192.168.56.60: icmp_seq=1 ttl=64 time=1.50 ms
64 bytes from 192.168.56.60: icmp_seq=2 ttl=64 time=0.779 ms
64 bytes from 192.168.56.60: icmp_seq=3 ttl=64 time=1.03 ms
64 bytes from 192.168.56.60: icmp_seq=4 ttl=64 time=0.675 ms
64 bytes from 192.168.56.60: icmp_seq=5 ttl=64 time=1.12 ms
64 bytes from 192.168.56.60: icmp_seq=6 ttl=64 time=0.706 ms
64 bytes from 192.168.56.60: icmp_seq=7 ttl=64 time=0.857 ms
64 bytes from 192.168.56.60: icmp_seq=8 ttl=64 time=1.95 ms
64 bytes from 192.168.56.60: icmp_seq=9 ttl=64 time=0.596 ms
64 bytes from 192.168.56.60: icmp_seq=10 ttl=64 time=0.595 ms
64 bytes from 192.168.56.60: icmp_seq=11 ttl=64 time=0.892 ms
64 bytes from 192.168.56.60: icmp_seq=12 ttl=64 time=0.765 ms
^C
--- 192.168.56.60 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11022ms
rtt min/avg/max/mdev = 0.595/0.955/1.952/0.387 ms
msantos@ubuntuServer:~$
```

UbuntuClient ping to the UbuntuServer



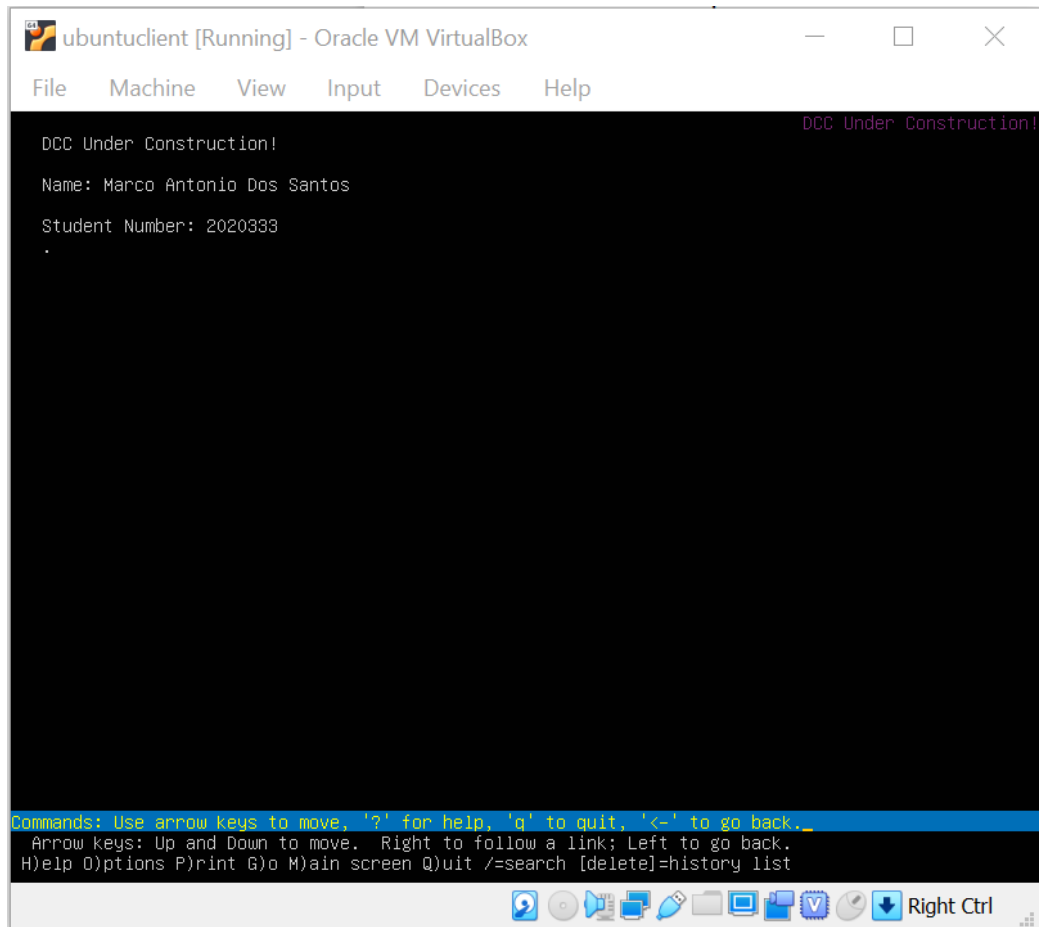
```
ubuntuclient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

RX packets 452 bytes 548145 (548.1 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 224 bytes 16014 (16.0 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

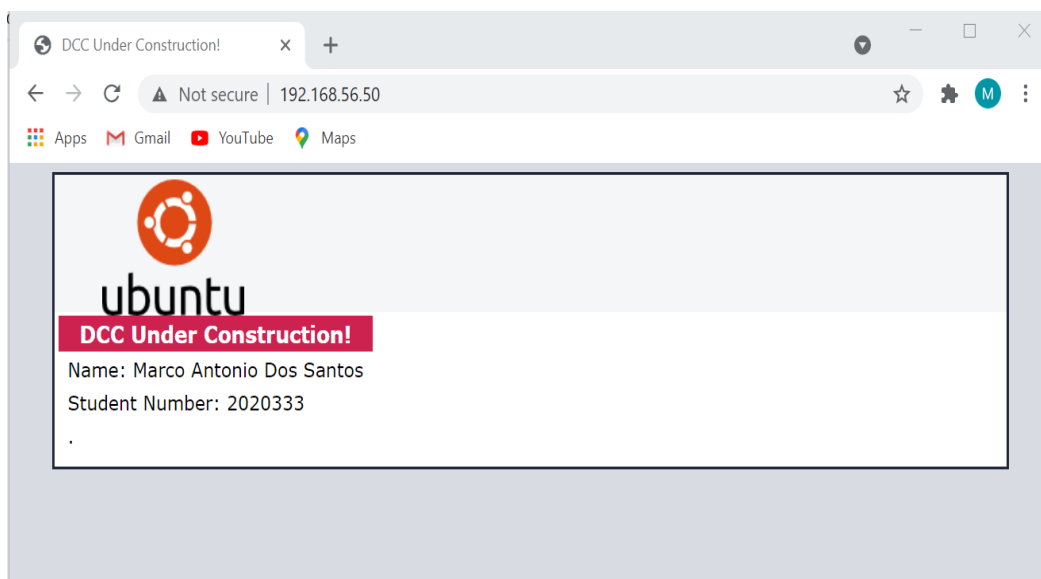
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 96 bytes 7444 (7.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 96 bytes 7444 (7.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msantos@ubuntuclient:~$ ping 192.168.56.50
PING 192.168.56.50 (192.168.56.50) 56(84) bytes of data:
64 bytes from 192.168.56.50: icmp_seq=1 ttl=64 time=0.421 ms
64 bytes from 192.168.56.50: icmp_seq=2 ttl=64 time=1.04 ms
64 bytes from 192.168.56.50: icmp_seq=3 ttl=64 time=0.974 ms
64 bytes from 192.168.56.50: icmp_seq=4 ttl=64 time=1.02 ms
64 bytes from 192.168.56.50: icmp_seq=5 ttl=64 time=1.00 ms
64 bytes from 192.168.56.50: icmp_seq=6 ttl=64 time=1.01 ms
64 bytes from 192.168.56.50: icmp_seq=7 ttl=64 time=0.996 ms
64 bytes from 192.168.56.50: icmp_seq=8 ttl=64 time=1.04 ms
64 bytes from 192.168.56.50: icmp_seq=9 ttl=64 time=1.02 ms
64 bytes from 192.168.56.50: icmp_seq=10 ttl=64 time=1.05 ms
64 bytes from 192.168.56.50: icmp_seq=11 ttl=64 time=1.13 ms
64 bytes from 192.168.56.50: icmp_seq=12 ttl=64 time=1.02 ms
64 bytes from 192.168.56.50: icmp_seq=13 ttl=64 time=1.02 ms
64 bytes from 192.168.56.50: icmp_seq=14 ttl=64 time=1.07 ms
64 bytes from 192.168.56.50: icmp_seq=15 ttl=64 time=0.968 ms
64 bytes from 192.168.56.50: icmp_seq=16 ttl=64 time=1.06 ms
^C
--- 192.168.56.50 ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15027ms
rtt min/avg/max/mdev = 0.421/0.989/1.126/0.151 ms
msantos@ubuntuclient:~$
```

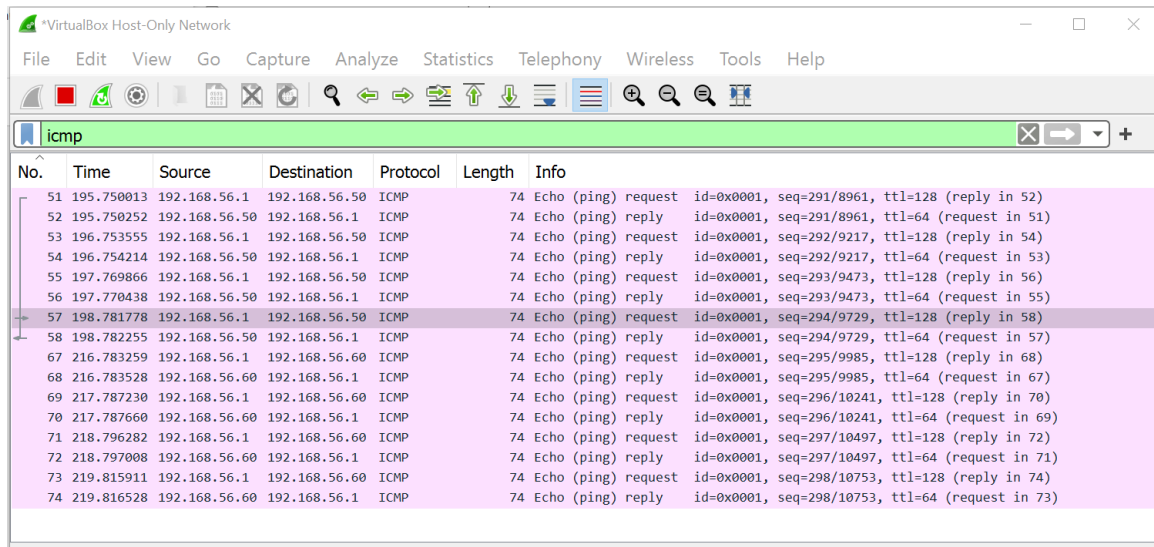
Test access to Apache home page!



Host operating system to access to the Apache home page



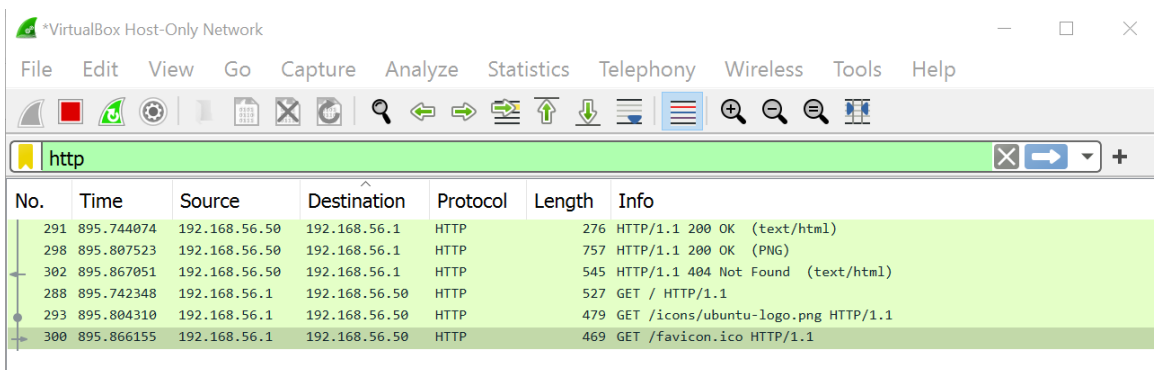
Wireshark capture of Icmp traffic



The image shows a Wireshark capture of ICMP traffic in a VirtualBox Host-Only Network. The capture is filtered for 'icmp'. The packet list shows a series of ping requests and replies between 192.168.56.1 and 192.168.56.50. The packet details pane shows the structure of an ICMP Echo (ping) request and reply, including fields like ID, sequence number, and TTL.

No.	Time	Source	Destination	Protocol	Length	Info
51	195.750013	192.168.56.1	192.168.56.50	ICMP	74	Echo (ping) request id=0x0001, seq=291/8961, ttl=128 (reply in 52)
52	195.750252	192.168.56.50	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=291/8961, ttl=64 (request in 51)
53	196.753555	192.168.56.1	192.168.56.50	ICMP	74	Echo (ping) request id=0x0001, seq=292/9217, ttl=128 (reply in 54)
54	196.754214	192.168.56.50	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=292/9217, ttl=64 (request in 53)
55	197.769866	192.168.56.1	192.168.56.50	ICMP	74	Echo (ping) request id=0x0001, seq=293/9473, ttl=128 (reply in 56)
56	197.770438	192.168.56.50	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=293/9473, ttl=64 (request in 55)
57	198.781778	192.168.56.1	192.168.56.50	ICMP	74	Echo (ping) request id=0x0001, seq=294/9729, ttl=128 (reply in 58)
58	198.782255	192.168.56.50	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=294/9729, ttl=64 (request in 57)
67	216.783259	192.168.56.1	192.168.56.60	ICMP	74	Echo (ping) request id=0x0001, seq=295/9985, ttl=128 (reply in 68)
68	216.783528	192.168.56.60	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=295/9985, ttl=64 (request in 67)
69	217.787230	192.168.56.1	192.168.56.60	ICMP	74	Echo (ping) request id=0x0001, seq=296/10241, ttl=128 (reply in 70)
70	217.787660	192.168.56.60	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=296/10241, ttl=64 (request in 69)
71	218.796282	192.168.56.1	192.168.56.60	ICMP	74	Echo (ping) request id=0x0001, seq=297/10497, ttl=128 (reply in 72)
72	218.797008	192.168.56.60	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=297/10497, ttl=64 (request in 71)
73	219.815911	192.168.56.1	192.168.56.60	ICMP	74	Echo (ping) request id=0x0001, seq=298/10753, ttl=128 (reply in 74)
74	219.816528	192.168.56.60	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=298/10753, ttl=64 (request in 73)

HTTP contents being transferred between servers



The image shows a Wireshark capture of HTTP traffic in a VirtualBox Host-Only Network. The capture is filtered for 'http'. The packet list shows several HTTP requests and responses between 192.168.56.1 and 192.168.56.50. The packet details pane shows the structure of an HTTP GET request and response, including fields like status code, content type, and content length.

No.	Time	Source	Destination	Protocol	Length	Info
291	895.744074	192.168.56.50	192.168.56.1	HTTP	276	HTTP/1.1 200 OK (text/html)
298	895.807523	192.168.56.50	192.168.56.1	HTTP	757	HTTP/1.1 200 OK (PNG)
302	895.867051	192.168.56.50	192.168.56.1	HTTP	545	HTTP/1.1 404 Not Found (text/html)
288	895.742348	192.168.56.1	192.168.56.50	HTTP	527	GET / HTTP/1.1
293	895.804310	192.168.56.1	192.168.56.50	HTTP	479	GET /icons/ubuntu-logo.png HTTP/1.1
300	895.866155	192.168.56.1	192.168.56.50	HTTP	469	GET /favicon.ico HTTP/1.1

2- SSH logging and encrypted traffic

Logging into ubuntuuser using SSH

```
msantos@ubuntuuser: ~  
login as: msantos  
msantos@192.168.56.50's password:  
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-65-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/advantage  
  
System information as of Sat  8 May 11:34:40 UTC 2021  
  
System load:  0.01          Processes:              116  
Usage of /:   50.0% of 8.79GB Users logged in:          1  
Memory usage: 24%          IPv4 address for enp0s3: 192.168.56.50  
Swap usage:   0%           IPv4 address for enp0s8: 10.0.3.15  
  
* Pure upstream Kubernetes 1.21, smallest, simplest cluster ops!  
  
https://microk8s.io/  
  
*** System restart required ***  
Last login: Sat May  8 11:32:24 2021 from 192.168.56.1  
msantos@ubuntuuser:~$
```

SSH encrypted Traffic

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.56.50	192.168.56.1	SSH	182	Server: Encrypted packet (len=128)
2	0.040843	192.168.56.1	192.168.56.50	TCP	54	52596 → 22 [ACK] Seq=1 Ack=129 Win=8209 Len=0
3	1.014683	192.168.56.50	192.168.56.1	SSH	166	Server: Encrypted packet (len=112)
4	1.055102	192.168.56.1	192.168.56.50	TCP	54	52596 → 22 [ACK] Seq=1 Ack=241 Win=8209 Len=0
5	2.016412	192.168.56.50	192.168.56.1	SSH	166	Server: Encrypted packet (len=112)
6	2.056893	192.168.56.1	192.168.56.50	TCP	54	52596 → 22 [ACK] Seq=1 Ack=353 Win=8208 Len=0
7	2.373791	192.168.56.1	239.255.255.250	SSDP	171	M-SEARCH * HTTP/1.1
8	2.395066	192.168.56.1	239.255.255.250	SSDP	166	M-SEARCH * HTTP/1.1
9	3.017464	192.168.56.50	192.168.56.1	SSH	182	Server: Encrypted packet (len=128)
10	3.057231	192.168.56.1	192.168.56.50	TCP	54	52596 → 22 [ACK] Seq=1 Ack=481 Win=8208 Len=0
11	4.022642	192.168.56.50	192.168.56.1	SSH	182	Server: Encrypted packet (len=128)

3-Rename Hostname(s)

Ubuntuserers rename

```
root@ubuntuserver:/home/msantos# cat /etc/hostname
ubuntuserver
root@ubuntuserver:/home/msantos# hostnamectl set-hostname web-server-333
root@ubuntuserver:/home/msantos# hostname
web-server-333
root@ubuntuserver:/home/msantos# hostnamectl
  Static hostname: web-server-333
            Icon name: computer-vm
            Chassis: vm
            Machine ID: 40621680576c401a8827ef4d7fbeb67
            Boot ID: f0ca8fbbb88f4b2193c1abc871dfe6c0
            Virtualization: oracle
            Operating System: Ubuntu 20.04.2 LTS
            Kernel: Linux 5.4.0-65-generic
            Architecture: x86-64
root@ubuntuserver:/home/msantos# hostname
web-server-333
```

Ubuntuclient rename

```
root@ubuntuclient:/home/msantos# cat /etc/hostname
ubuntuclient
root@ubuntuclient:/home/msantos# hostnamectl set-hostname web-client-333
root@ubuntuclient:/home/msantos# hostname
web-client-333
root@ubuntuclient:/home/msantos# hostnamectl
  Static hostname: web-client-333
            Icon name: computer-vm
            Chassis: vm
            Machine ID: 9a0687241d3e4aecadb9047e9e74d4cc
            Boot ID: b87e49cf7ae84c60ab1eaa98ae091950
            Virtualization: oracle
            Operating System: Ubuntu 20.04.2 LTS
            Kernel: Linux 5.4.0-72-generic
            Architecture: x86-64
root@ubuntuclient:/home/msantos# hostname
web-client-333
```

Setting the Ip address permanent

The directory and file was created but unfortunately it was not possible to set ip address permanent of both Ubuntu servers ,look at images below:

Ubuntuserver:

```
GNU nano 4.8 /etc/netplan/00-installer-config.yaml Modified
# This is the network config written by 'subiquity'
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s3:
      dhcp4: no
      dhcp6: no
      addresses: [192.168.56.100/24]
      gateway4: 192.168.56.1
      nameservers:
        addresses: [8.8.8.8, 1.1.1.1]
```

```
msantos@web-server-333:~$ sudo netplan try
Error while loading /etc/netplan/00-installer-config.yaml, aborting.
msantos@web-server-333:~$ sudo netplan apply
/etc/netplan/00-installer-config.yaml:10:5: Invalid YAML: inconsistent indentation:
  gateway4: 192.168.56.1
msantos@web-server-333:~$ _
```

As you can see in this ubuntu server tell us a error in gateway4.

Ubuntu client

```
GNU nano 4.8 /etc/netplan/00-installer-config.yaml Modified
# This is the network config written by 'subiquity'
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      dhcp8: no
      addresses: [192.168.56.200/24]
      gateway4: 192.168.56.1
      nameservers:
        addresses: [8.8.8.8, 1.1.1.1]
```

```
msantos@web-client-333:~$ sudo netplan apply
/etc/netplan/00-installer-config.yaml:8:5: Error in network definition: unknown key 'dhcp8'
  dhcp8: no
  ^
msantos@web-client-333:~$ sudo netplan try
/etc/netplan/00-installer-config.yaml:8:5: Error in network definition: unknown key 'dhcp8'
  dhcp8: no
  ^

An error occurred: the configuration could not be generated
Reverting.
Warning: Stopping systemd-networkd.service, but it can still be activated by:
  systemd-networkd.socket
msantos@web-client-333:~$
```

In these Ubuntu server tell us that error in dhcp8, both server were using the same proceeding however gave a few different errors, maybe they were caused by spaces errors or even typing.

4- Firewall

SSH not permitted

```
msantos@web-server-333:~$ sudo ufw deny in ssh
[sudo] password for msantos:
Skipping adding existing rule
Skipping adding existing rule (v6)
msantos@web-server-333:~$ sudo ufw status
Status: active

To Action From
--
22/tcp DENY Anywhere
22/tcp (v6) DENY Anywhere (v6)
```

HTTP traffic not being permitted

```
msantos@web-server-333:~$ sudo ufw deny out http
Rule added
Rule added (v6)
msantos@web-server-333:~$ sudo ufw status
Status: active

To Action From
--
22/tcp DENY Anywhere
80/tcp DENY Anywhere
22/tcp (v6) DENY Anywhere (v6)
80/tcp (v6) DENY Anywhere (v6)

80/tcp DENY OUT Anywhere
80/tcp (v6) DENY OUT Anywhere (v6)

msantos@web-server-333:~$
```

SSH being allow and other traffic not permitted

```
msantos@web-server-333:~$ sudo ufw status
Status: active

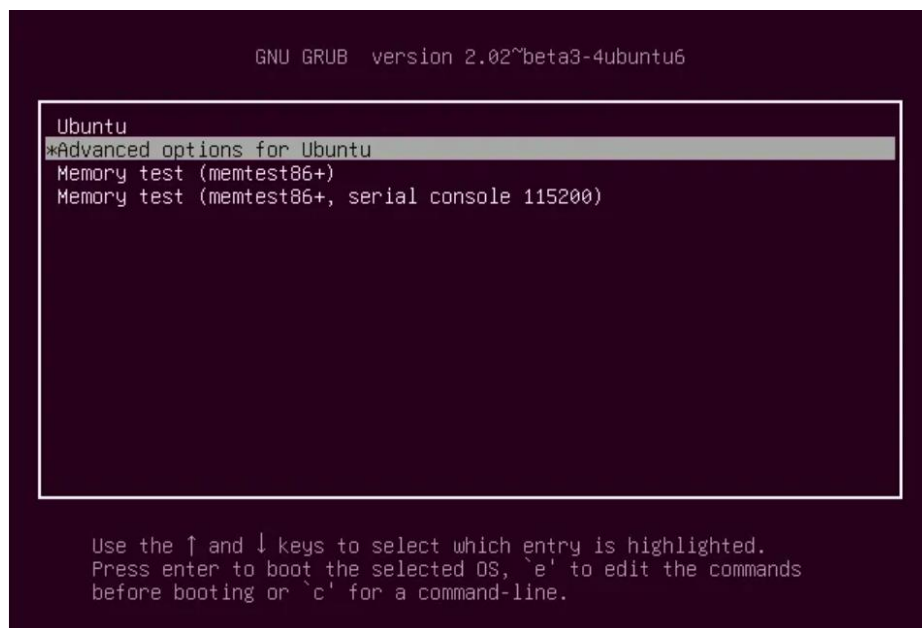
To Action From
--
22/tcp ALLOW Anywhere
80/tcp DENY Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) DENY Anywhere (v6)

80/tcp DENY OUT Anywhere
80 DENY OUT Anywhere
80/tcp (v6) DENY OUT Anywhere (v6)
80 (v6) DENY OUT Anywhere (v6)

msantos@web-server-333:~$ _
```

5-How to recover the Password Of Ubuntu Linux

It's possible easily redefine reset a user's password in Ubuntu using a few commands at system startup. It must initiate Ubuntu machine and press Esc Key continuously after the bios screen to open the Ubuntu grub menu , as shown below. Then, select "Advanced option for Ubuntu" and press the Enter key.



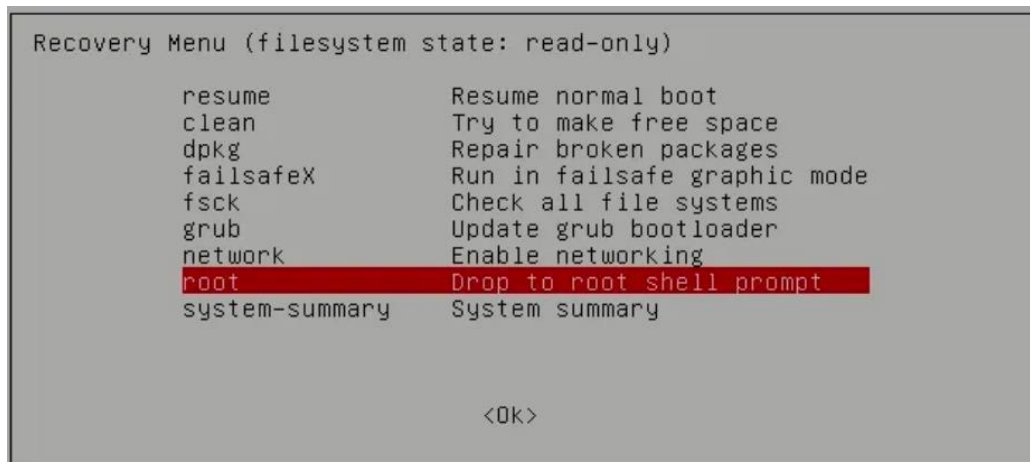
Source: It's foss

The **recovery Mode** option will appear. Press it to continue the process:



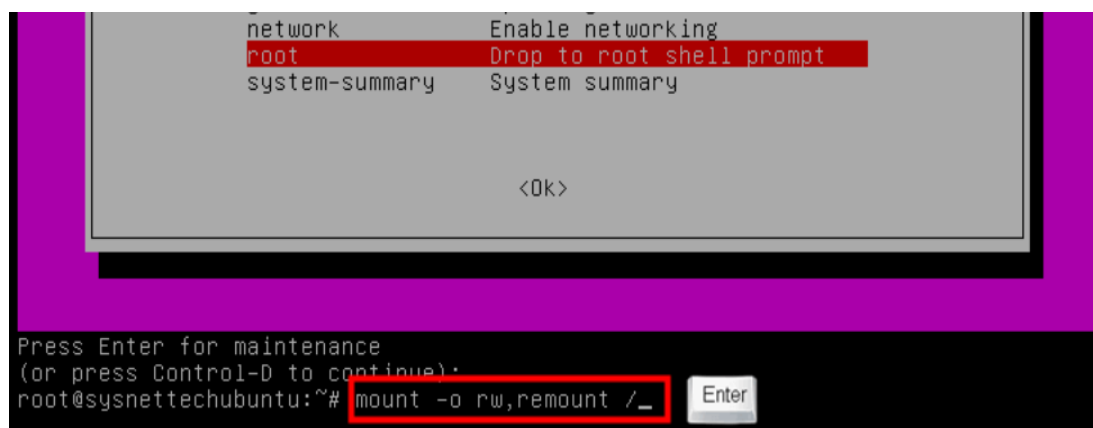
Source: It's foss

In **Recovery Mode** choose the option root-drop to root shell prompt:



Source: It's foss

When Selectin the **root – drop to root shell prompt** option in recovery mode, the command line will be displayed look at image below. Soon after, it will be necessary to reassemble the read- only file system for write permission it important by executing the command below:



Source: It's foss

Ready! It is possible now to run the password recovery command look at how suppose to be shown on screen:

```
root@sysnettechubuntu:~# passwd (username)
```

Source: It's foss

Set the new password and complete the operation by executing the exit command to return to the recovery menu, look at the image below. Then press the **Resume** option:

```
Recovery Menu (filesystem state: read-only)

resume      Resume normal boot
clean       Try to make free space
dpkg        Repair broken packages
failsafeX   Run in failsafe graphic mode
fsck        Check all file systems
grub        Update grub bootloader
network     Enable networking
root        Drop to root shell prompt
system-summary  System summary

<Ok>
```

Source: It's foss

When prompted for the user's password on the login screen it must enter the new password registered in this operation!

Reference:

Prakash,A.,2021.How to Reset Ubuntu Password in 2 minutes .[online] available at:
<https://itsfoss.com/how-to-hack-ubuntu-password/> ,[Accessed 06 may 2021].